

Statistical Foundations of Machine Learning

Practice Set 2

1 Homework problems

Note: solutions to problems in this section are to be submitted and will be graded. Deadline for submission is Wed, 11th March. Total marks : 15. Justify your answers and present all calculations/arguments clearly.

1. Teams A and B play a ‘best-of-five’ series with the first team to win three games wins the series (the series ends immediately when such an event happens and no further games are played then). A wins a game with probability p independently of the result of the other games.

Let X_k be indicator variable for A winning the series in k games. Note that for any event $\mathcal{E}$, the indicator random variable $1_{\mathcal{E}}$ takes two values: 1 if $\mathcal{E}$ is true (i.e. with probability $\mathbb{P}(\mathcal{E})$) and 0 otherwise.

- (a) (2 marks) For X_k , determine its probability mass function (pmf) as a function of p .
 - (b) (2 marks) Let X be the indicator variable that A wins the series. Determine the pmf of X .
 - (c) (2 marks) If A has won the first game, determine the conditional pmf of X and evaluate it for $p = 0.5$.
2. A random variable X is said to have the $\text{Laplace}(a, b)$ distribution if it has the following pdf with constants $b > 0$, a , and C .

$$f_X(x) = C \exp(-b|x - a|), \quad x \in \mathbb{R}.$$

- (a) (1 mark) Plot the function for, say, $a = 1, b = 2$.
 - (b) (1 mark) Determine C in terms of a and b that would make this a valid density function.
 - (c) (2 marks) Find the cumulative distribution function (CDF) of X .
 - (d) (2 marks) $X \sim \text{Laplace}(a, 1)$ and $Y \sim \text{Laplace}(0, 1)$ are two independent random variables. Find $\text{Prob}(X > Y)$ for i) $a > 0$ and ii) $a < 0$.
3. Let $X \sim \text{Bin}(n, p)$ be a Binomial random variable. Let Y be a random variable that has a binomial distribution conditional on X , i.e., $\{Y|X\}$ is $\text{Bin}(X, q)$.
- (a) (1 mark) Determine the range of values that Y can take.
 - (b) (2 marks) For m in the range of Y , obtain the conditional PMF of Y given $X = k$, i.e., obtain $\text{Prob}(Y = m|X = k)$ for different values of m . Note this probability will be a function of m and k .

2 Practice problems

Note: problems in this section are for practice, solutions not to be submitted

1. Let X be a discrete random variable with the following PMF -

$$P_X(x) = \begin{cases} 0.1 & x = 1, 2, 10 \\ 0.3 & x = 5 \\ 0.4 & x = 8 \\ 0 & \text{otherwise} \end{cases}$$

Find and plot the CDF of X .

2. Buses arrive at a bus-stop at 10 minute intervals starting at noon. A man arrives at the bus-stop at a random time X minutes after noon, where X has the CDF:

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{60} & 0 \leq x \leq 60 \\ 1 & x > 60 \end{cases}$$

What is the probability that the man waits less than 5 minutes for a bus?

3. I roll a fair die repeatedly until a number larger than 4 is observed. If N is the total number of times that I roll the die, find $P(N = k)$, for $k = 1, 2, 3, \dots$.
4. There are two jointly continuous random variables X and Y with the joint pdf-

$$f_{XY}(x, y) = \begin{cases} \frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{6} & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

For $0 \leq y \leq 2$, find:

- (a) The conditional PDF of X given $Y = y$.
- (b) $P(X < \frac{1}{2} | Y = y)$
5. Two discrete random variables x and y have the given joint distribution. Find the following ($F_{xy}(x, y)$ represents the joint CDF) :

	x=1	x=2	x=3
y=1	3/12	1/12	3/12
y=2	1/12	3/12	1/12

- (a) $F_{xy}(1, 1)$
- (b) $F_{xy}(1.5, 4)$
- (c) $F_{xy}(4, 9)$

(d) $F_{xy}(1, -1.5)$

6. Romeo and Juliet live in cities R and J but work in cities J and R respectively. Each day they start for work at a uniformly random time between 8 am and 9:30 am independent of each other. Both of them travel at the same constant speed and it takes 25 minutes for each of them to reach the other city. What is the probability that Romeo and Juliet meet each other on their way to work on some particular day?

7. The random variables X, Y have joint PDF $f_{X,Y}(x, y)$:

$$f_{X,Y}(x, y) = \begin{cases} cx(y-x)e^{-y} & 0 \leq x \leq y \\ 0 & \text{elsewhere} \end{cases},$$

where c is a constant.

- (a) Compute the value of c .
 (b) Obtain the conditional density function $f_{X|Y}(x|y)$.
 8. The random variable X has the following CDF.

$$F_X(x) = \begin{cases} x & \text{for } 0 \leq x < 1/3 \\ \frac{3x+1}{6} & \text{for } 1/3 \leq x < 2/3 \\ x & \text{for } 2/3 \leq x < 1 \\ 1 & \text{for } x \geq 1 \end{cases}$$

- (a) What is $\mathbb{P} X < 1/2$?
 (b) What is $\mathbb{P} X < 2/3$?
 (c) Is X a continuous random variable?
 9. Consider two independent random variables $X_1 \sim \text{Exp}(\lambda_1)$ and $X_2 \sim \text{Exp}(\lambda_2)$. Find the probability $\mathbb{P} X_1 \leq X_2$.

$$X \sim \text{Exp}(\lambda): f_X(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

10. Let N be a geometric random variable with PMF

$$p_N(k) = p(1-p)^{k-1}, k = 1, 2, \dots, p \in (0, 1).$$

Find the conditional PMF $p_N(k|A) = \mathbb{P}\{N = k|N \in A\}$ if $A = \{N \text{ is even}\}$.

11. **CDF & PDF** The probability density function (PDF) of a random variable has the following form.

$$f(x) = \begin{cases} a & \text{for } x < 0 \\ x & \text{for } x \in [0, 1] \\ b(1-x) & \text{for } x \in (1, 2] \\ c & \text{for } x > 2 \end{cases}$$

Here, a, b, c are constants.

- (a) What are the values of a , b and c ?
- (b) Compute the CDF corresponding to this PDF.
12. Suppose that n tubelights in a room are turned on at time 0. The lifetimes of the tubelights are independent and are exponentially distributed with parameter μ .
- (a) What is the cdf of the time after which the first tubelight fuses out?
- (b) Find the cdf of the time after which the room goes dark?

13. The number of pages X in a document has PMF

$$P_X(x) = \begin{cases} 0.15 & \text{if } x = 1, 2, 3, 4 \\ 0.1 & \text{if } x = 5, 6, 7, 8 \\ 0 & \text{otherwise} \end{cases}$$

A firm sends all documents with an even number of pages to printer A and all documents with an odd number of pages to printer B .

- (a) Find the conditional PMF of the length X of a document, given the document was sent to printer A .
- (b) Find the conditional PMF of the length X of a document, given the document was sent to printer B and had no more than six pages.
14. A positive integer is selected, with $P(I = n) = \frac{1}{2^n}$, $n = 1, 2, \dots$. If I takes the value n , a coin with probability of heads equal to e^{-n} is tossed once. Find the probability that the resulting toss will be a head.
15. Let $f(x, y) = cx^2y$ for $x^2 \leq y \leq 1$ denote the joint pdf for (X, Y) . Find:
- (a) c
- (b) $\mathbb{P}[|X| \geq Y]$
- (c) Marginals $f_X(x)$ and $f_Y(y)$
16. X is a strictly positive discrete random variable, i.e., it takes values in the set $\{1, 2, \dots\}$ and $p_X(x)$ is its pmf. If the pmf has the form $p_X(x) = \frac{C \cdot 2^x}{x!}$, find the value of C . Repeat for $p_X(x) = \frac{C \cdot 2^{-x}}{x}$, and $p_X(x) = Cx^\alpha$.
17. For the previous problem, find (i) $\text{Prob}(X > 2)$, (ii) most probable value of X .
18. There are n coins each of which comes up heads with probability p . In round 1, each coin is tossed once. In round 2, all those coins that came up heads in round 1 are tossed again and others are left unchanged. Let X be the total number of heads after round 2. Find the pmf of X .
19. Let X be the random variable denoting the number of tosses of a biased coin to get the first head. The first head has not occurred in the first k tosses. What is the probability that you will have to toss the coin n more times.